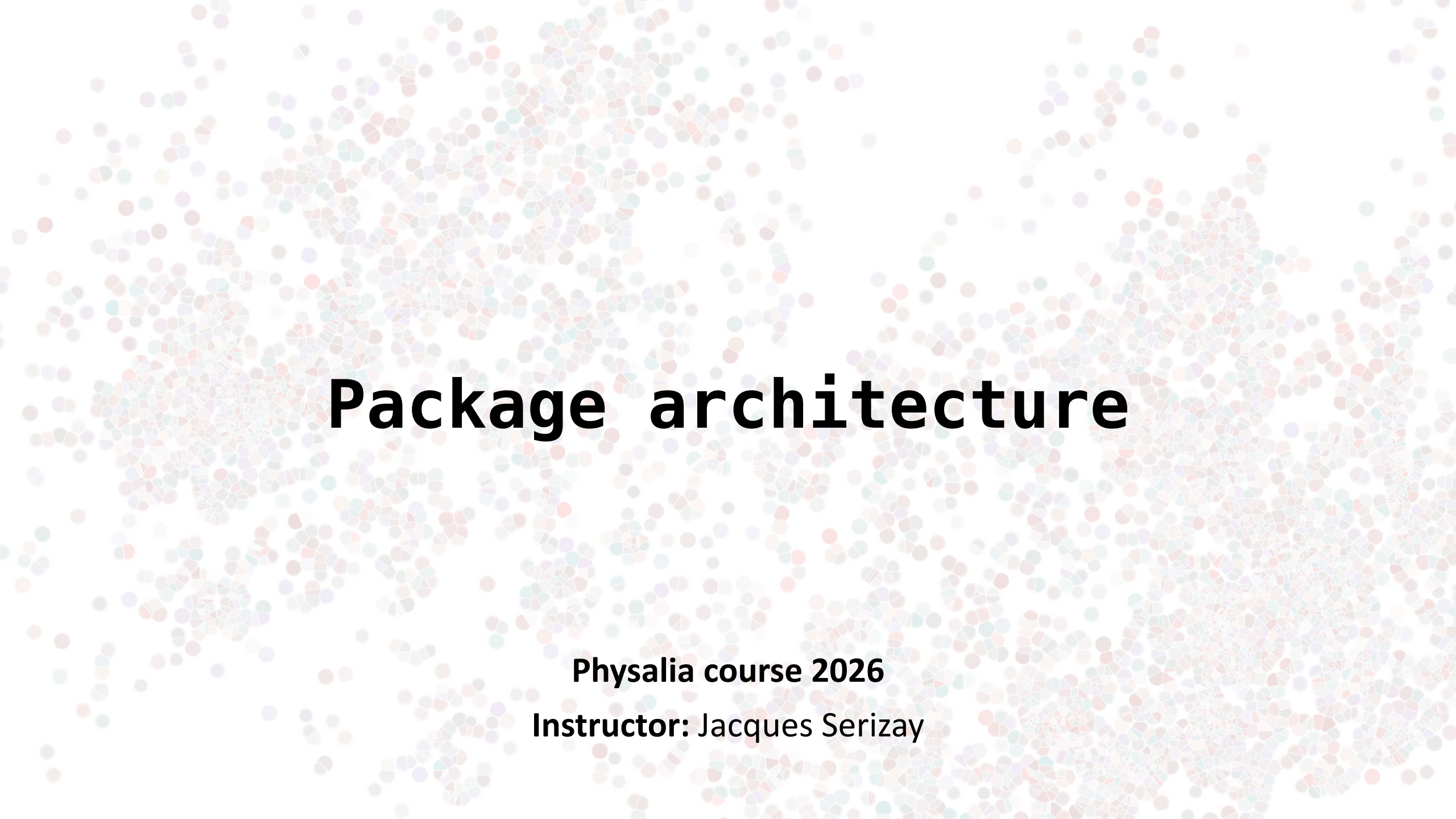




Package architecture

Physalia course 2026

Instructor: Jacques Serizay

What are packages

- R “core” is already a mix of several packages !
 - base
 - compiler
 - datasets
 - graphics
 - grDevices
 - grid
 - methods
 - parallel
 - splines
 - stats
 - stats4
 - tcltk
 - tools
 - translations
 - utils

What are packages

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 - compiler
 - datasets
 - graphics
 - grDevices
 - grid
 - methods
 - parallel
 - splines
 - stats
 - stats4
 - tcltk
 - tools
 - translations
 - utils

R preferred extension mechanism: new functionalities (through new functions, but not only!)

What are packages

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- base
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- stats
- stats4
- tcltk
- tools
- translations
- utils

30424

CRAN PACKAGES

2418

BIOCONDUCTOR PACKAGES

2241

R-FORGE PACKAGES

74409

GITHUB PACKAGES

R preferred extension mechanism: new functionalities (through new functions, but not only!)

What you may know

```
myProject/
```

```
  functions.R
```

```
  utils.R
```

Standard package source: the bare minimum

```
myPackage/
```

```
R/
```

```
  functions.R
```

```
  utils.R
```

```
DESCRIPTION
```

```
NAMESPACE
```

Standard package source: more complete structure

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Package: HiContacts
Title: HiContacts: R interface to cool files
Version: 0.99.11
Date: 2022-08-16
Authors@R:
 c(person(given = "Jacques",
 family = "Serizay",
 role = c("aut", "cre"),
 email = "jacquesserizay@gmail.com",
 comment = c(ORCID = "0000-0002-4295-0624")))
Description: HiContacts: R interface to (m)cool files and other Hi-C
 processed file formats. HiContacts provides a collection of tools to
 analyse and visualize Hi-C datasets. It can import data from pairs
 or (m)cool files.
License: MIT + file LICENSE
URL: <https://github.com/js2264/HiContacts>
BugReports: <https://github.com/js2264/HiContacts/issues>
Depends:
 R (>= 4.2)
Imports:
 HiContactsData
Suggests:
 rmarkdown
biocViews:
 HiC
Encoding: UTF-8
VignetteBuilder: knitr
LazyData: false
Roxygen: list(markdown = TRUE)
RoxygenNote: 7.2.1

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Package ‘HiContacts’

October 14, 2022

Title HiContacts: R interface to cool files
Version 0.99.11
Date 2022-08-16
Description HiContacts: R interface to (m)cool files and other Hi-C processed file formats. HiContacts provides a collection of tools to analyse and visualize Hi-C datasets. It can import data from pairs or (m)cool files.
License MIT + file LICENSE
URL <https://github.com/js2264/HiContacts>
BugReports <https://github.com/js2264/HiContacts/issues>
Depends R (>= 4.2)
Imports HiContactsData, InteractionSet, GenomicInteractions, GenomicRanges, IRanges, GenomeInfoDb, S4Vectors, BiocGenerics, methods, rhdf5, Matrix, vroom, tibble, tidyr, dplyr, glue, stringr, reticulate, ggplot2, ggrastr, scales
Suggests cowplot, testthat (>= 3.0.0), BiocStyle, knitr, rmarkdown
biocViews HiC, DNA3DStructure, DataImport
Encoding UTF-8
VignetteBuilder knitr
LazyData false
Roxygen list(markdown = TRUE)
RoxygenNote 7.2.1
git_url <https://git.bioconductor.org/packages/HiContacts>
git_branch master
git_last_commit 6ccb53e
git_last_commit_date 2022-10-13
Date/Publication 2022-10-14
Author Jacques Serizay [aut, cre] (<<https://orcid.org/0000-0002-4295-0624>>)
Maintainer Jacques Serizay <jacquesserizay@gmail.com>

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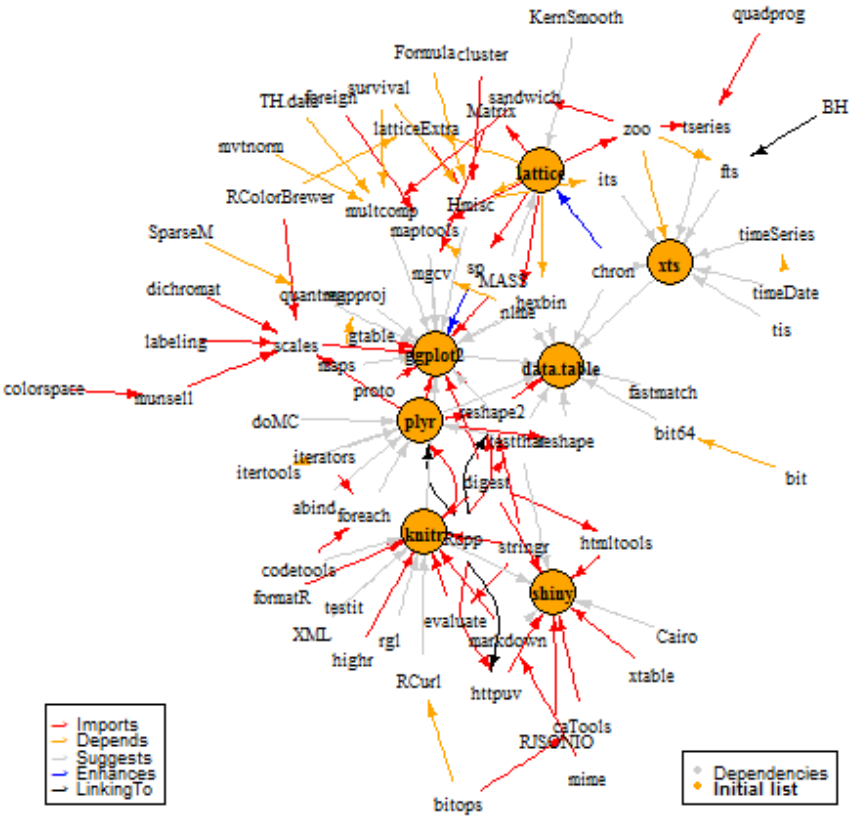
NEWS

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Generated by roxygen2: do not edit by hand

```
export(APA)
export(cool2gi)
export(cool2seqinfo)
export(detrend)
export(fetchCool)
export(formatCoords)
export(getAnchors)
export(getCounts)
export(getHicStats)
export(getPs)
export(ggMatrix)
export(ggPs)
export(ggtheme_HiContacts)
export(ggtheme_HiContacts_tracks)
export(gi2cm)
export(lsCoolFiles)
export(lsCoolResolutions)
export(pairs2gi)
export(peekCool)
export(plotMatrix)
export(splitCoords)
import(BiocParallel)
import(InteractionSet)
import(dplyr)
import(ggplot2)
import(ggrastr)
import(rhdf5)
import(scales)
import(stringr)
import(tibble)
import(tidyr)
import(tools)
import(zeallot)
importFrom(GenomeInfoDb,Seqinfo)
importFrom(GenomeInfoDb,seqlengths)
importFrom(GenomicRanges,GRanges)
importFrom(GenomicRanges,end)
importFrom(GenomicRanges,findOverlaps)
importFrom(GenomicRanges,mcols)
importFrom(GenomicRanges,resize)
importFrom(GenomicRanges,seqnames)
importFrom(GenomicRanges,start)
```

Package dependency graph



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HiContacts

HiContacts provides tools to import `(m)cool` matrices in R and work with them there.

It creates a new `Contacts` class of objects, built on pre-existing Bioconductor objects, namely `InteractionSet`, `GenomicInteractions` and `ContactMatrix` (Lun, Perry & Ing-Simmons, F1000Research 2016), and provides **analytical** and **visualization** tools to investigate contact maps.

Installation

`HiContacts` will soon be available in Bioconductor. To install the current release use:

```
```\nif (!requireNamespace("BiocManager", quietly = TRUE))\ninstall.packages("BiocManager")\nBiocManager::install("HiContacts")\n```\n
```

For now, `HiContacts` can be installed from Github as follows:

```
```\ninstall.packages("devtools")\ndevtools::install_github("js2264/HiContacts")\nlibrary(HiContacts)\n```\n
```

Citation

If you are using `HiContacts` in your research, please cite:

```
> Serizay J (2022). _HiContacts: HiContacts: R interface to cool files_.\n> R package version 0.99.0\n> <https://github.com/js2264/HiContacts>.\n
```

How to use `HiContacts`

`HiContacts` includes a introduction vignette where its usage is illustrated. To access the vignette, please use:

```
```\nvignette('HiContacts')\n```\n
```

## Overview

### Import a `(m)cool` file as `Contacts`

## HiContacts

lifecycleexperimentalR-CMD-check-biocpassingpkgdownpassing

HiContacts provides tools to import `(m)cool` matrices in R and work with them there.

It creates a new `Contacts` class of objects, built on pre-existing Bioconductor objects, namely `InteractionSet`, `GenomicInteractions` and `ContactMatrix` (Lun, Perry & Ing-Simmons, F1000Research 2016), and provides **analytical** and **visualization** tools to investigate contact maps.

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Serizay J (2022). HiContacts: HiContacts: R interface to cool files. R package version 0.99.0\nhttps://github.com/js2264/HiContacts.\n
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### How to use `HiContacts`

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```
vignette('HiContacts')\n
```

### Overview

Import a `(m)cool` file as `Contacts`

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```
periodicDNA 0.3.2 (dev)

* IMPORTANT:
- Implemented data-raw for reproducibility

* MINOR:
- Changed xlim of norm. distr. plot in plotPeriodicityResults()

periodicDNA 0.3.1 (2020-05-05)

* IMPORTANT:
- rollmean(k=3) is now applied *before* normalisation *as well*,
 on the raw distribution vector
- plotPeriodicityResults() output returns one single plot (with cowplot)
- getPeriodicityTrack() now returns the Rle
- Improved plotting functions -now show shuffled for plotPeriodicityResults()
- Added ggplot2 theming

* MINOR:
* Changed many variable names (all to snake_case)
* sampleGRanges is now full-fledged function
 (GRanges, DNASTringSet, character and BSgenome methods)
* sampleGenome is an alias for sampleGRanges.character
* Added sacCer3 to getPeriodicity BSgenomes
* Added DNASTring method for getPeriodicity
* Added a vignette describing the internal steps
* Clarified user-level functions in README
* Added ce11_TSSs data
* Renamed generateperiodicitytrack as getPeriodicityTrack
* Renamed variables in getFPI and getPeriodicity
* Created a utility char2BSgenome()

periodicDNA 0.3.0 (2020-05-03)

* Added tests
* Added getFPI function
* cleaned-up functions
* cleaned-up function dependencies
* Added toy data
* Added vignette

periodicDNA 0.2.1 (2020-03-04)

* Added Travis build check
* Simplified README.md
```

# LICENSE

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=====

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**## Preamble**

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For the developers' and authors' protection, the GPL clearly explains that there is no warranty for this free software. For both users' and authors' sake, the GPL requires that modified versions be marked as changed, so that their problems will not be attributed erroneously to authors of previous versions.

Some devices are designed to deny users access to install or run modified versions of the software inside them. although the manufacturer can do so. This is fundamentally

# Function files

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**functions.R**

utils.R

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```
#' Parsing (m)cool files
#'
#' These functions are the workhorse internal functions used to import
#' a `.(m)cool` file as GenomicInteractions (wrapped into a `Contacts` object
#' by `Contacts()` function).
#'
#' @param file file
#' @param resolution resolution
#' @param balanced import balancing scores
#' @return anchors from (m)cool, stored as a GRanges
#'
#' @importFrom GenomicRanges GRanges
#' @importFrom GenomicRanges seqnames
#' @importFrom GenomicRanges start
#' @importFrom GenomicRanges end
#' @importFrom IRanges IRanges
#' @rdname parse

getAnchors <- function(file, resolution = NULL, balanced = "cooler") {
 bins <- fetchCool(file, "bins", resolution)
 anchors <- GenomicRanges::GRanges(
 bins$chr,
 IRanges::IRanges(bins$start + 1, bins$end),
 bin_id = seq_along(bins$chr),
 seqinfo = cool2seqinfo(file, resolution)
)
 names(anchors) <- paste(GenomicRanges::seqnames(anchors), GenomicRanges::start(anchors), GenomicRanges::end(anchors), sep
= " ")
 if (~"weight" %in% names(bins) & {
 balanced == "cooler" | balanced == TRUE
 }) {
 anchors$weight <- as.numeric(bins$weight)
 } else {
 weight <- 1
 }
 return(anchors)
}
```

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```
% Generated by roxygen2: do not edit by hand
% Please edit documentation in R/parse.R
\name{getAnchors}
\alias{getAnchors}
\alias{getCountsFromPair}
\alias{getCounts}
\alias{fetchCool}
\alias{lsCoolFiles}
\alias{lsCoolResolutions}
\alias{peekCool}
\alias{cool2seqinfo}
\alias{cool2gi}
\alias{gi2cm}
\alias{cm2matrix}
\alias{pairs2gi}
\title{Parsing (m)cool files}
\usage{
 getAnchors(file, resolution = NULL, balanced = "cooler")
}
\arguments{
 \item{file}{pairs file: \verb{<readname>\\t<chr1>\\t<start1>\\t<chr2>\\t<start2>}}

 \item{resolution}{resolution}

 \item{balanced}{import balancing scores}
}
\value{
 anchors from (m)cool, stored as a GRanges
}
\description{
 These functions are the workhorse internal functions used to import
 a \verb{.(m)cool} file as GenomicInteractions (wrapped into a \code{Contacts} object
 by \code{Contacts()} function).
}
```

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pdf documentation once package is published

---

getAnchors	<i>Parsing (m)cool files</i>
------------	------------------------------

---

**Description**

These functions are the workhorse internal functions used to import a .(m)cool file as GenomicInteractions (wrapped into a Contacts object by Contacts() function).

**Usage**

getAnchors(file, resolution = NULL, balanced = "cooler")

**Arguments**

file	pairs file: <readname>\t<chr1>\t<start1>\t<chr2>\t<start2>
resolution	resolution
balanced	import balancing scores

**Value**

anchors from (m)cool, stored as a GRanges



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?getAnchors

getAnchorspackage:HiContactsR Documentation

[Parsing \(m\)cool files](#)[Draw](#)[Design](#)[Transitions](#)[Animations](#)[Slide](#)

Description:

These functions are the workhorse internal functions used to import a .(m)cool file as GenomicInteractions (wrapped into a 'Contacts' object by 'Contacts()'  
function).

Usage:

```
getAnchors(file, resolution = NULL, balanced = "cooler")
```

Arguments:

file: pairs file: <readname>\t<chr1>\t<start1>\t<chr2>\t<start2>

resolution: resolution

balanced: import balancing scores

Value:

anchors from (m)cool, stored as a GRanges

# Source/Bundle/Binary/Installed/Loaded: késako?

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Source

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r-lib / devtoolsPublic

<> CodeIssues12Pull requests1ActionsProjectsWikiSecurityInsights

main40 branches48 tagsGo to fileAdd fileCode

jennybcIncrement version number to 2.4.5.9000aa3f88b23 days ago3,463 commits

.github	Add linting action (#2431)	5 months ago
R	Set package in test_active_file() (#2471)	last month
inst	spelling::update_wordlist()	4 months ago
man-roxygen	Clarify documentation of the pkg argument	3 years ago
man	Surface unused imports with check(cran = TRUE) (#2461)	3 months ago
pkgdown/favicon	Minor improvements to website (#2256)	2 years ago
revdep	Check revdeps	26 days ago
tests	Set package in test_active_file() (#2471)	last month
vignettes	Remove last vestiges of the master branch	12 months ago
.Rbuildignore	Rbuildignore submission file	4 months ago
.gitattributes	enable union merge for NEWS.md file	9 years ago
.gitignore	Ignore some stuff	4 months ago
.lintr	Add linting action (#2431)	5 months ago
DESCRIPTION	Increment version number to 2.4.5.9000	23 days ago
LICENSE	Re-license as MIT (#2334)	2 years ago
LICENSE.md	Re-license as MIT (#2334)	2 years ago
MAINTENANCE.md	Add maintenance document	12 months ago
NAMESPACE	Quiet check (#2447)	4 months ago
NEWS.md	Increment version number to 2.4.5.9000	23 days ago
README.md	Update cheatsheet links	4 months ago
_pkgdown.yml	Switch to pkgdown "auto" development mode	11 months ago
codecov.yml	Convert to using GitHub Actions	3 years ago
cran-comments.md	Check revdeps	26 days ago
devtools.Rproj	RStudio update	5 months ago

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devtools.r-lib.orgpackage-creation

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Releases35

devtools 2.4.5Latest23 days ago

+ 34 releases

Packages

No packages published

Contributors145

+ 134 contributors

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Languages

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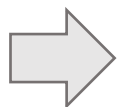
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myPackage.tar.gz

**Bundle**

## Source/Bundle/ Binary /Installed/Loaded: késako?

```
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```

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utils.R

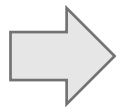
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myPackage.tar.

r-lib / devtoolsPublic

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main40 branches48 tagsGo to fileAdd fileCode

jennybcIncrement version number to 2.4.5.9000aa3f88b23 days ago3,463 commits

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pkgdown/favicon	Minor improvements to website (#2256)	2 years ago
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.gitattributes	enable union merge for NEWS.md file	9 years ago
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DESCRIPTION	Increment version number to 2.4.5.9000	23 days ago
LICENSE	Re-license as MIT (#2334)	2 years ago
LICENSE.md	Re-license as MIT (#2334)	2 years ago
MAINTENANCE.md	Add maintenance document	12 months ago
NAMESPACE	Quiet check (#2447)	4 months ago
NEWS.md	Increment version number to 2.4.5.9000	23 days ago
README.md	Update cheatsheet links	4 months ago
_pkgdown.yml	Switch to pkgdown "auto" development mode	11 months ago
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AboutTools to make an R developer's life easier

devtools.r-lib.orgpackage-creation

ReadmeUnknown, MIT licenses foundCode of conduct2.2k stars127 watching750 forks

Releases35

devtools 2.4.5Latest23 days ago

+ 34 releases

Packages

No packages published

Contributors145

+ 134 contributors

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github-pagesActive

# Source/Bundle/ Binary /Installed/Loaded: késako?

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23 days ago  
jennybc  
v2.4.5  
4a7ba32  
Compare

devtools 2.4.5 Latest

- `check(cleanup =)` was deprecated in devtools v1.11.0 (2016-04-12) and was made defunct in v2.4.4 (2022-07-20). The documentation is more clear now about recommended alternatives.
- `check(check_dir = NULL)` is the new default, to align with the default behaviour of the underlying `rcmdcheck::rcmdcheck()`.
- `check(cran = TRUE)` sets the env var `_R_CHECK_PACKAGES_USED_IGNORE_UNUSED_IMPORTS_` to `FALSE`, in order to surface the "Namespace in Imports field not imported from" NOTE. This only applies to R >= 4.2, due to favorable changes in the behaviour of R CMD check --as-cran (#2459).
- `test_active_file()` passes the package onto to testthat so it can correctly set the `TESTTHAT_PKG` envvar (#2470).
- `test()` and `test_active_file()` once again work with testthat itself.

▼ Assets 2

Source code (zip) 23 days ago

Source code (tar.gz) 23 days ago

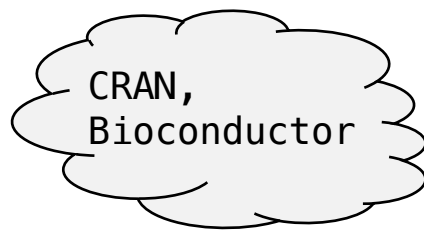
😊



Source/Bundle/ Binary /Installed/Loaded: késako?

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→ myPackage.tar.gz



Binary

Package Archives

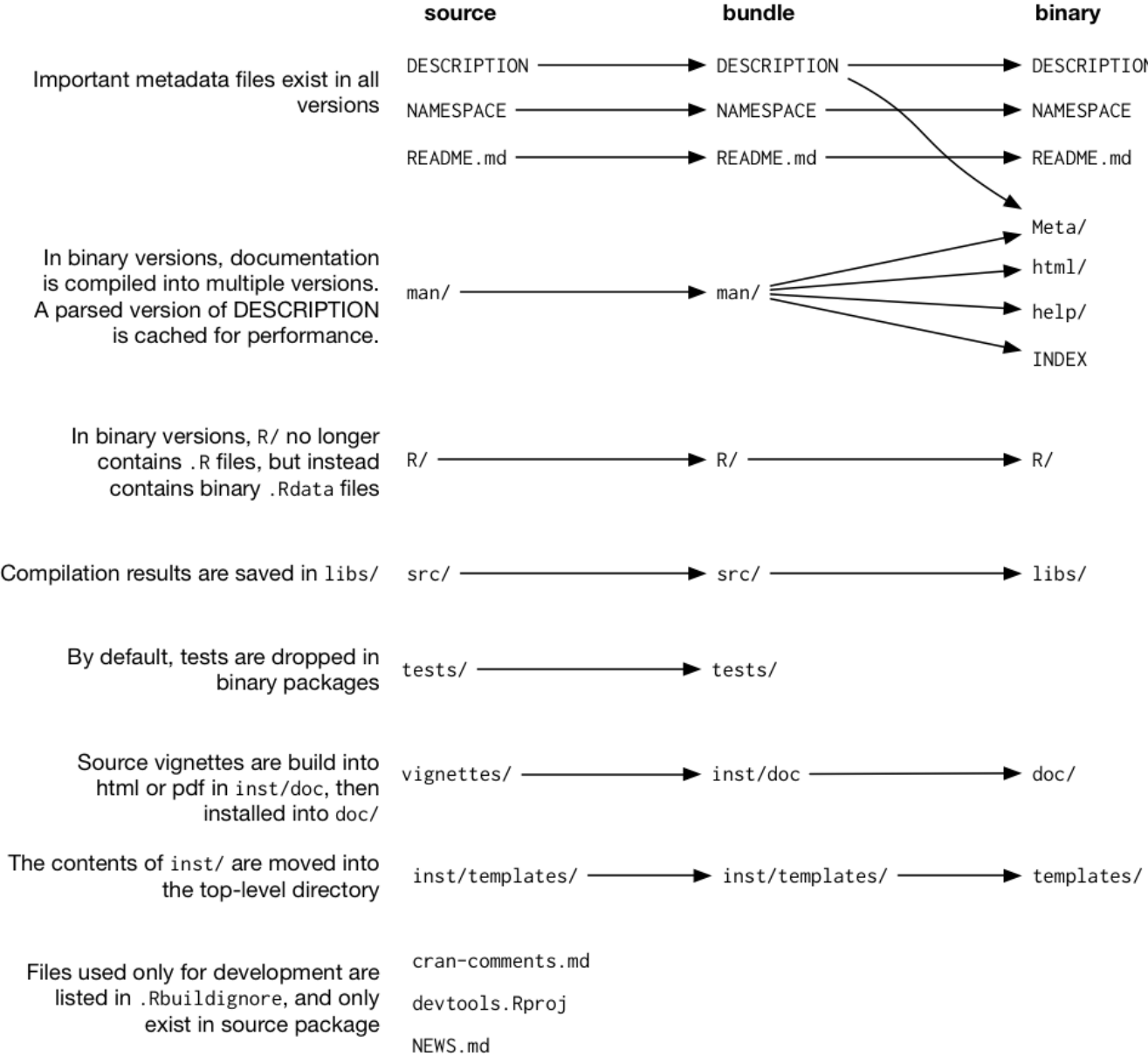
Follow [Installation](#) instructions to use this package in your R session.

Source Package	<a href="#">S4Vectors 0.34.0.tar.gz</a>
Windows Binary	<a href="#">S4Vectors 0.34.0.zip</a>
macOS Binary (x86_64)	<a href="#">S4Vectors 0.34.0.tgz</a>

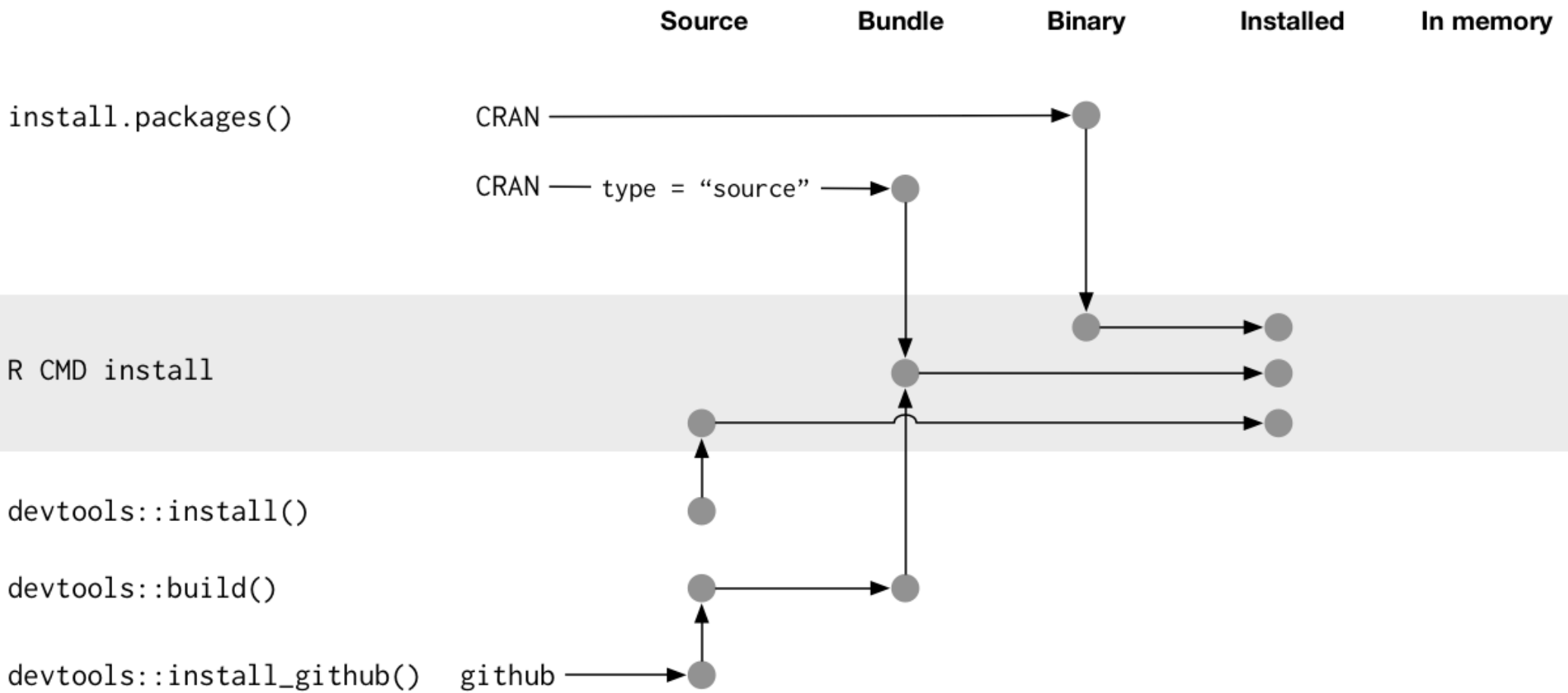




# Source/Bundle/ Binary /Installed/Loaded: késako?



# Source/Bundle/ Binary /Installed/Loaded: késako?



Read more...

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<https://r-pkgs.org/Structure.html>